Group Cohomology

Shing Tak Lam

Lent 2023*

Contents

1 **Definitions and resolutions** 1

2 **Low degree cohomology, group extensions** 10

Lecture 1

1 Definitions and resolutions

Let G be a group.

Definition 1.1 (integral group ring)
The *integral group ring* $\mathbb{Z}G$ has elements being formal sums

$$\sum n_g g$$

for $n_g \in \mathbb{Z}, g \in G$, and only finitely many n_g are non-zero. This is a free abelian group under addition, i.e.

$$\left(\sum m_g g\right) + \left(\sum n_g g\right) = \sum (m_g + n_g) g$$

and we have multiplication

$$\left(\sum m_h h\right)\left(\sum n_k k\right) = \sum_g \left(\sum_{hk=g} m_h n_k\right) g$$

This is the associative ring underlying the integral representation theory of G. We will write $1 = 1e$ for the multiplicative identity in G.

Definition 1.2 (module)
A *(left)* $\mathbb{Z}G$*-module* M is an abelian group under addition, and a map

$$\begin{aligned} \mathbb{Z}G \times M &\to M \\ (r, m) &\mapsto rm \end{aligned}$$

such that

1. $r(m+n) = rm + rn$,
2. $(r_1 + r_2)m = r_1 m + r_2 m$,
3. $(r_1 r_2)m = r_1(r_2 m)$,

*Based on lectures by Chris Brookes. Last updated January 30, 2024.

4. $1m = m$

A module is *trivial* if $gm = m$ for all $g \in G$, $m \in M$. We say *the trivial module* is $\mathbb{Z}$ with trivial action $gm = m$. A *free* $\mathbb{Z}G$*-module* on X is formal sums

$$\sum r_x x$$

for $r_x \in \mathbb{Z}G$, $x \in X$, and finitely many r_x non-zero. We will write $\mathbb{Z}G\{X\}$ for this.

Submodules and *quotient modules* are defined as we would expect.

Definition 1.3 ($\mathbb{Z}G$-map)

An $\mathbb{Z}G$*-map (or morphism)* $\alpha : M_1 \to M_2$ is a map of abelian groups, with

$$\alpha(rm_1) = r\alpha(m_1)$$

for $r \in \mathbb{Z}G$, $m_1 \in M_1$.

Example 1.4

The *augmentation map* is

$$\begin{aligned} \varepsilon : \mathbb{Z}G &\to \mathbb{Z} \\ \sum n_g g &\mapsto \sum n_g \end{aligned}$$

where we treat $\mathbb{Z}G$ as a left $\mathbb{Z}G$-module, and $\mathbb{Z}$ as the trivial module. This is a $\mathbb{Z}G$ map, and it is also a ring homomorphism.

Notation 1.5. Write $\mathrm{Hom}_G(M, N)$ for the set of $\mathbb{Z}G$-maps from $M \to N$, with pointwise addition.

Example 1.6

If we regard $\mathbb{Z}G$ as a left $\mathbb{Z}G$-module, then

$$\mathrm{Hom}_G(\mathbb{Z}G, M) \cong M$$

for any left $\mathbb{Z}G$-module M, by sending φ to $\varphi(1)$. This is an isomorphism since

$$\varphi(r) = \varphi(r1) = r\varphi(1)$$

Note that $\mathrm{Hom}_G(\mathbb{Z}G, M)$ is a left $\mathbb{Z}G$-module, with

$$(s\phi)(r) = \phi(rs)$$

In particular,

$$\mathrm{Hom}_G(\mathbb{Z}G, \mathbb{Z}G) \cong \mathbb{Z}G$$

With this, $\varphi \mapsto \varphi(1)$ corresponds to multiplication on the *right* by $\varphi(1)$.

Note that G may not be abelian, so the distinction between left and right modules matters.

Definition 1.7

If $f : M_1 \to M_2$ is a $\mathbb{Z}G$-map, then the dual map is

$$f^* : \mathrm{Hom}_G(M_2, N) \to \mathrm{Hom}_G(M_1, N)$$

for any $\mathbb{Z}G$-module N. This is given by

$$f^*(\varphi) = \varphi \circ f$$

Similarly, if $f : N_1 \to N_2$, then we have

$$f_* : \mathrm{Hom}_G(M, N_1) \to \mathrm{Hom}(M, N_2)$$

with $f_*(\varphi) = f \circ \varphi$.

These are maps of abelian groups, since in general $\mathrm{Hom}_G(M, N)$ doesn't have to be a $\mathbb{Z}G$-module, as we don't have a right action.

Example 1.8 (Prototypical example)

Let $G = \langle t \rangle$ be an infinite cyclic group, and consider the graph Γ with vertices $\{v_i\}_{i\in\mathbb{Z}}$, edges $v_i \leftrightarrow v_{i+1}$. Then G acts on the set V of vertices, by $tv_i = v_{i+1}$, and also on the set E of edges. The action is transitive in both cases. The formal integral sums $\mathbb{Z}V$ and $\mathbb{Z}E$ are $\mathbb{Z}G$-modules, and are free. Fix the edge $e : v_0 \leftrightarrow v_1$. Then we have $\mathbb{Z}G$-maps

$$\begin{aligned} \mathrm{d} : \mathbb{Z}E &\to \mathbb{Z}V \\ e &\mapsto v_1 - v_0 \end{aligned}$$

and also

$$\begin{aligned} \mathbb{Z}V &\to \mathbb{Z} \\ v_0 &\mapsto 1 \end{aligned}$$

This correspond to the augmentation map.

Definition 1.9 (chain complex, exact, homology)

A *chain complex* of $\mathbb{Z}G$-modules is a sequence

$$M_s \xrightarrow{\mathrm{d}_s} M_{s-1} \xrightarrow{\mathrm{d}_{s-1}} \cdots \xrightarrow{\mathrm{d}_{t+1}} M_t$$

of $\mathbb{Z}G$-modules and maps, such that for $t < n < s$, $\mathrm{d}_n \circ \mathrm{d}_{n+1} = 0$. We will write

$$M_\bullet = (M_n, \mathrm{d}_n)_{t\le n\le s}$$

We say that $M_\bullet$ is *exact* at M_n if $\mathrm{im}(\mathrm{d}_{n+1}) = \ker(\mathrm{d}n)$. The sequence is exact if if it exact at all M_n with $t < n < s$.

The *homology* of $M_\bullet$ is given by

$$\begin{aligned} \mathrm{H}_s(M_\bullet) &= \ker(\mathrm{d}s) \\ \mathrm{H}_n(M_\bullet) &= \frac{\ker(\mathrm{d}_n)}{\mathrm{im}(\mathrm{d}_{n+1})} \text{ for } t < n < s \\ \mathrm{H}_t(M_\bullet) &= \mathrm{coker}(\mathrm{d}_{t+1}) \end{aligned}$$

A *short exact sequence* is a chain complex of the form

$$0 \longrightarrow M_1 \xrightarrow{\alpha} M_2 \xrightarrow{\beta} M_3 \longrightarrow 0$$

In our prototypical example, we have a short exact sequence

$$0 \longrightarrow \mathbb{Z}E \longrightarrow \mathbb{Z}V \longrightarrow \mathbb{Z} \longrightarrow 0$$

which corresponds to

$$0 \longrightarrow \mathbb{Z}G \longrightarrow \mathbb{Z}G \longrightarrow \mathbb{Z} \longrightarrow 0$$

where the map $\mathbb{Z}G \to \mathbb{Z}G$ is multiplication on the right by $t - 1$.

Definition 1.10 (projective)

A $\mathbb{Z}G$-module P is *projective* if for every surjective $\mathbb{Z}G$-map $\alpha : M_1 \to M_2$, and every $\mathbb{Z}G$-map $\beta : P \to M_2$, there exists $\overline{\beta}$ such that

$$\begin{array}{ccccc} & & & & P \\ & & \overset{\overline{\beta}}{\nearrow} & & \downarrow{\scriptstyle \beta} \\ M_1 & \xrightarrow[\alpha]{} & M_2 & \longrightarrow & 0 \end{array}$$

commutes.

If we have any short exact sequence

$$0 \longrightarrow N \xrightarrow{f} M_1 \xrightarrow{\alpha} M_2 \longrightarrow 0$$

Consider

$$0 \longrightarrow \mathrm{Hom}_G(P, N) \xrightarrow{f_*} \mathrm{Hom}_G(P, M_1) \xrightarrow{\alpha_*} \mathrm{Hom}_G(P, M_2) \longrightarrow 0 \qquad (*)$$

Then P is projective if and only if $(*)$ is exact. Note that in $(*)$, we always have exactness except at $\mathrm{Hom}_G(P, M_2)$.

Lecture 2

Lemma 1.11. Free modules are projective.

Proof. Let $\alpha : M_1 \to M_2$ be a surjective $\mathbb{Z}G$-map, $\beta : \mathbb{Z}G\{X\} \to M_2$. For each $x \in X$, since α is surjective, there exists $m_x \in M_1$ such that $\alpha(m_x) = \beta(x)$. We can then define $\overline{\beta} : \mathbb{Z}G\{X\} \to M_1$, by

$$\sum r_x \cdot x \mapsto \sum r_x \cdot m_x$$

□

Definition 1.12

A *projective (resp. free) resolution of the trivial module* $\mathbb{Z}$ is an exact sequence

$$\cdots \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} \mathbb{Z} \longrightarrow 0$$

with all the P_i projective (resp. free).

Example 1.13

When $G = \langle t \rangle$ is infinite cyclic, we have the free resolution

$$0 \longrightarrow \mathbb{Z}G \xrightarrow{\cdot(t-1)} \mathbb{Z}G \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

Example 1.14

If $G = \langle t \rangle$ cyclic of order n. Then

$$\cdots \longrightarrow \mathbb{Z}G \xrightarrow{\alpha} \mathbb{Z}G \xrightarrow{\beta} \mathbb{Z}G \xrightarrow{\alpha} \mathbb{Z}G \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

is a free resolution, where $\alpha(x) = x(t-1)$ and $\beta(x) = x(1 + t + \cdots + t^{n-1})$.

From algebraic topology, if we have a connected simplicial complex X, with $\pi_1(X) = G$, and the universal cover $\widetilde{X}$ is contractible, then we have a free resolution of $\mathbb{Z}$. The point is that the simplicial complex X contains a lot of information about G. We are trying to replicate this algebraically.

For calculation purposes, we're interested in *small resolutions*. For example, where the free modules have small rank. But for theory development purposes, we're wanting general constructions, and such resolutions tend to be large.

Definition 1.15

G is of type FP_n if $\mathbb{Z}$ has a projective resolution

$$\cdots \longrightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} \mathbb{Z} \longrightarrow 0$$

so that $P_n, \ldots, P_0$ are finitely generated.

G is of type FP_n is $\mathbb{Z}$ has a projective resolution with all P_n finitely generated.

G is of type FP if there exists a projective resolution of $\mathbb{Z}$ with finite length and all P_n are finitely generated. That is, $P_s = 0$ for s sufficiently large.

Example 1.16

$G = \langle t \rangle$ infinite cyclic is of type FP, $G = C_n = \langle t \rangle$ finite cyclic is of type FP_∞.

These are to be regarded as finiteness conditions on the group. The topological version of FP_n (which is called F_n) would be asking for X with $\pi_1(X) = G$, having a finite n-skeleton. Note that these two conditions are not equivalent.

Now lets meet some general constructions. If we have a partial projective resolution

$$P_s \xrightarrow{d_s} P_{s-1} \longrightarrow \cdots \longrightarrow P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} \mathbb{Z} \longrightarrow 0$$

Then we can make it longer by setting

$$P_{s+1} = \mathbb{Z}G\{X_{s+1}\}$$

with $X_{s+1} = \ker(d_s)$, and set

$$d_{s+1} : \sum r_x x \mapsto \sum r_x x$$

Then

$$P_{s+1} \xrightarrow{d_{s+1}} P_s \xrightarrow{d_s} P_{s-1} \longrightarrow \cdots \longrightarrow P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} \mathbb{Z} \longrightarrow 0$$

is a longer partial projective resolution. To make P_{s+1} smaller, we could take X_{s+1} to be a $\mathbb{Z}G$-generating set for $\ker(d_s)$. This is particularly useful if $\ker(d_s)$ is finitely generated.

Continuing this, we can get a (full) projective resolution.

Definition 1.17

The *standard (or bar) resolution of* $\mathbb{Z}$, for any group G, is: Let

$$G^{(n)} = \{[g_1|\ldots|g_n] \mid g_i \in G\}$$

be the set of symbols. In particular, $G^{(0)} = \{[]\}$ Let

$$F_n = \mathbb{Z}G\{G^{(n)}\}$$

be the free module on $G^{(n)}$. The differential is given by

$$\begin{aligned} d_n[g_1|\ldots|g_n] = {} & g_1[g_2|\ldots|g_n] \\ & -[g_1g_2|g_3|\ldots|g_n] \\ & +[g_1|g_2g_3|\ldots|g_n] + \ldots \\ & +(-1)^{n-1}[g_1|\ldots|g_{n-1}g_n] \\ & +[g_1|\ldots|g_{n-1}] \end{aligned}$$

With this, $d_{n-1}d_n = 0$, and we have a chain complex

$$\cdots \longrightarrow F_n \longrightarrow F_{n-1} \longrightarrow \cdots \longrightarrow F_0 \xrightarrow{[]\mapsto 1} \mathbb{Z} \longrightarrow 0$$

Remark 1.18. The bar resolution corresponds to the standard resolution in algebraic topology. Consider $n+1$ tuples G^{n+1}, and form the free abelian group $\mathbb{Z}G^{n+1}$ under addition. G acts on G^{n+1} diagonally, so

$$g \cdot (h_0, \dots, h_n) = (gh_0, \dots, gh_n)$$

So $\mathbb{Z}G^{n+1}$ is a free $\mathbb{Z}G$-module, on the basis of $n+1$ tuples with $g_0 = 1$. Then we have a correspondence

$$[g_1|\dots|g_n] \mapsto (1, g_1, g_1g_2, \dots, g_1 \cdots g_n)$$

Note that removing the first entry gives $g_1(1, g_2, g_2g_3, \dots, g_2 \cdots g_n)$, and removing the seconc entry in the tuple gives $(1, g_1g_2, g_1g_2g_3, \dots, g_1 \cdots g_n)$.

Lemma 1.19. The bar resolution is exact.

Proof. We'll just consider the d_n as maps of abelian groups. F_n has basis $G \times G^{(n)}$ as a free abelian group, and $G \times G^{(n)}$ is the set

$$\{g_0[g_1|\dots|g_n] \mid g_i \in G\}$$

Define $\mathbb{Z}$-linear maps $s_n : F_n \to F_{n+1}$, such that

$$\mathrm{id}_{F_n} = \mathrm{d}_{n+1}s_n + s_{n-1}\mathrm{d}_n \tag{$*$}$$

given by

$$s_n(g_0[g_1|\dots|g_n]) = [g_0|\dots|g_n]$$

We can verify that $(*)$ holds on the basis, and so if $x \in \ker(\mathrm{d}_{n+1})$, then

$$x = \mathrm{id}(x) = \mathrm{d}_{n+1}s_n(x) + s_{n-1}\mathrm{d}_n(x) = \mathrm{d}_{n+1}s_n(x) \in \mathrm{im}(\mathrm{d}_{n+1})$$

That is, s_n is a chain homotopy from the identity to zero. □

Corollary 1.20. A finite group is of type FP_∞.

Proof. The bar resolution gives such a resolution. □

Lecture 3

Definition 1.21

Consider a projective resolution

$$\cdots \longrightarrow P_{n+1} \longrightarrow P_n \xrightarrow{d_n} \cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow \mathbb{Z} \longrightarrow 0$$

of $\mathbb{Z}$ by $\mathbb{Z}G$-modules. Let M be a (left) $\mathbb{Z}G$-module. Apply the functor $\mathrm{Hom}_G(\cdot, M)$, we get the *dual sequence*

$$\cdots \longleftarrow \mathrm{Hom}(P_{n+1}, M) \xleftarrow{\mathrm{d}^{n+1}} \mathrm{Hom}(P_n, M) \longleftarrow \cdots \longleftarrow \mathrm{Hom}(P_0, M) \tag{$\dagger$}$$

where $\mathrm{d}^n = \mathrm{d}_n^*$. Then the *$n$-th cohomology group* $\mathrm{H}^n(G, M)$ *with coefficients in* M *is:*

$$\mathrm{H}^n(G, M) = \frac{\ker(\mathrm{d}^{n+1})}{\mathrm{im}(\mathrm{d}^n)} \text{ for } n \geq 1$$
$$\mathrm{H}^0(G, M) = \ker(\mathrm{d}^1)$$

Remark 1.22. We have dropped the $\mathbb{Z}$-term in $(\dagger)$.

Also, these cohomology groups are the homology groups of the chain complex

$$C_n = \mathrm{Hom}_G(P_{-n}, M)$$

with $-\infty < n \leq 0$.

Later, we will see that these cohomology groups are independent of the choice of resolution.

Example 1.23

In our prototypical example, $G = \langle t \rangle$ infinite cyclic, we have a projective resolution

$$0 \longrightarrow \mathbb{Z}G \xrightarrow{\mathrm{d}} \mathbb{Z}G \longrightarrow \mathbb{Z} \longrightarrow 0$$

where $\mathrm{d}(x) = x(t-1)$. For $\phi \in \mathrm{Hom}_G(\mathbb{Z}G, M)$, $x \in \mathbb{Z}G$,

$$\mathrm{d}^1(\phi)(x) = \phi(\mathrm{d}_1(x)) = \phi(x(t-1))$$

Moreover, recall $\mathrm{Hom}_G(\mathbb{Z}G, M) \cong M$, where $i(\phi) = \phi(1)$. In particular,

$$\mathrm{d}^1(\phi) \mapsto \mathrm{d}^1\phi(1) = \phi(t-1) = (t-1)\phi(1) = (t-1)i(\phi)$$

In this case, we have

$$0 \longleftarrow M \xleftarrow[(t-1)\cdot]{} M$$

and so

$$\mathrm{H}^0(G, M) = \{m \in M \mid tm = m\} = M^G$$
$$\mathrm{H}^1(G, M) = \frac{M}{(t-1)M} = M_G$$

and $\mathrm{H}^n(G, M) = 0$ otherwise. Here, M^G is the group of *invariants*, the largest submodule with trivial G-action. Conversely, M_G is the *coinvariants*, the largest quotient module with trivial G-action.

Remark 1.24. $\mathrm{H}^0(M, G) = M^G$ is generally true. But $\mathrm{H}^1(M, G) = M_G$ is special behaviour of $G = \langle t \rangle$.

For any group of type FP, we have that $\mathrm{H}^n(G, M) = 0$ for large enough n.

Definition 1.25

G is of *cohomological dimension m over $\mathbb{Z}$* if there exists an $\mathbb{Z}G$-module M, with

$$\mathrm{H}^m(G, M) \neq 0$$

and $\mathrm{H}^n(G, M') = 0$ for any $\mathbb{Z}G$-module M', and $n > m$.

Note that for all G, $\mathrm{H}^0(G, \mathbb{Z}) = \mathbb{Z} \neq 0$.

Example 1.26

$G = \langle t \rangle$ infinite cyclic is of cohomological dimension 1 (over $\mathbb{Z}$).

An exercise is to show that if G is a free group of finite rank, then it is also of cohomological dimension 1. To see this, the Cayley graph is a tree and we can construct a resultion using this.

Remark 1.27. The converse is true. A finitely generated group of cohomological dimension 1 is free. In fact, this is true in general. See [Stallings, 1968], Swan, 1969.

Now consider the bar resolution in our definition of cohomology. Note that

$$\mathrm{Hom}_G(\mathbb{Z}G\{G^{(n)}\}, M) \cong C^n(G, M) = \{\text{functions } G^{(n)} \to M\}$$

This is the same as function $G^n \to M$. We also have that

$$C^0(G, M) = \{\text{functions } [] \to M\} = M$$

Definition 1.28

The group of *n-cochains* of G with coefficients in M is $C^n(G, M)$. The *n-coboundary map*

$$\mathrm{d}^n : C^{n-1}(G, M) \to C^n(G, M)$$

is dual to d_n in the bar resolution. For $\phi \in C^{n-1}(G, M)$,

$$\begin{aligned}(\mathrm{d}^n\phi)(g_1, \ldots, g_n) = {} & g_1\phi(g_2, \ldots, g_n) \\ & - \phi(g_1g_2, \ldots, g_n) \\ & + \phi(g_1, g_2g_3, \ldots, g_n) \\ & + \ldots \\ & + (-1)^{n-1}\phi(g_1, g_2, \ldots, g_{n-1}g_n) \\ & + (-1)^n\phi(g_1, g_2, \ldots, g_{n-1})\end{aligned}$$

The group of *n-cocycles* is $Z^n(G, M) = \ker(\mathrm{d}^{n+1}) \subseteq C^n(G, M)$, and the *n-coboundaries* is $B^n(G, M) = \mathrm{im}(\mathrm{d}^n) \subseteq C^n(G, M)$.

Thus,

$$\mathrm{H}^n(G, M) = \frac{Z^n(G, M)}{B^n(G, M)}$$

Corollary 1.29. For any group G, $\mathrm{H}^0(G, M) = M^G$ is the invariants.

Definition 1.30

A *derivation of G with coefficients in M* is a function $\phi : G \to M$, such that

$$\phi(gh) = g\phi(h) + \phi(g)$$

Note that $Z^1(G, M)$ is the set of derivations. Also note that these are also called 'crossed homomorphisms'.

An *inner derivation* is ϕ of the form

$$\phi(g) = gm - m$$

for a fixed $m \in M$.

Corollary 1.31.

$$\mathrm{H}^1(G, M) = \frac{\{\text{derivations } G \to M\}}{\{\text{inner derivations} G \to M\}}$$

In particular, if M is a trivial $\mathbb{Z}G$-module, then

$$\mathrm{H}^1(G, M) = \mathrm{Hom}_{\mathrm{Group}}(G, M)$$

We will return to considering homology arising from different resolutions.

Definition 1.32

Let $(A_n, \alpha_n), (B_n, \beta_n)$ be chain complexes of $\mathbb{Z}G$-modules, a *chain map* $f = (f_n)$ are $\mathbb{Z}G$-maps $f_n : A_n \to B_n$,

such that

$$\begin{array}{ccc} A_n & \xrightarrow{\alpha_n} & A_{n-1} \\ {\scriptstyle f_n}\downarrow & & \downarrow{\scriptstyle f_{n-1}} \\ B_n & \xrightarrow[\beta_n]{} & B_{n-1} \end{array}$$

commutes.

Lemma 1.33. Given a chain map (f_n), it induces a map on homology groups

$$f_* : \mathrm{H}_n(A_\bullet) \to \mathrm{H}_n(B_\bullet)$$

Proof. Let $x \in \ker(\alpha_n)$. Define $f_*([x]) = [f_n(x)]$, where $[\cdot]$ denotes a homology class. Observe that

$$f_n(x) \in \ker(\beta_n)$$

since $\beta_n(f_n(x)) = f_{n-1}(\alpha_n(x)) = 0$. Moreover, if $x' = x + \alpha_{n-1}(y)$, then

$$f_n(x') = f_n(x) + f_n(\alpha_{n-1}(y)) = f_n(x) + \beta_n(f_n(y))$$

and so $[f_n(x')] = [f_n(x)]$. Moreover, this gives a map of abelian groups. □

Theorem 1.34. The definition of $\mathrm{H}^n(G, M)$ is independent of the choice of projective resolution.

Proof. Take projective resolutions (P_n, d_n) and (P'_n, d'_n) of $\mathbb{Z}$ by $\mathbb{Z}G$-modules. We will produce

- chain maps $f_n : P_n \to P'_n$, $g_n : P'_n \to P_n$,
- chain homotopies $s_n : P_n \to P_{n+1}$, $s'_n : P'_n \to P'_{n+1}$,

That is,

$$\mathrm{d}s + s\mathrm{d} = gf - \mathrm{id} \quad \text{and} \quad \mathrm{d}s' + s'\mathrm{d} = fg - \mathrm{id}$$

Using this data, the f_n define chain maps

$$f^* : \mathrm{Hom}_G(P'_n, M) \to \mathrm{Hom}_G(P_n, M) \quad \text{and} \quad g^* : \mathrm{Hom}_G(P'_n, M) \to \mathrm{Hom}_G(P_n, M)$$

which induce maps between the respective homology groups. Now observe if $\phi \in \ker(\mathrm{d}^{n+1}) \in \mathrm{Hom}(P_n, M)$, then

$$\begin{aligned} f^*g^*(\phi)(x) &= \phi(g(f(x))) \\ &= \phi(x) + \phi(\mathrm{d}s(x)) + \phi(s\,\mathrm{d}(x)) \\ &= \phi(x) + s^*\mathrm{d}\phi(x) + \mathrm{d}s^*\phi(x) \\ &= \phi(x) + \mathrm{d}s^*\phi(x) \end{aligned}$$

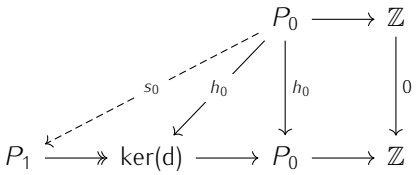
and so $f^*g^*(\phi) = \phi + \mathrm{d}(s^*\phi)$. Hence f^*g^* induces the identity map on homology. Similarly, g^*f^* also induce the identity map. This, f^*, g^* yield isomorphisms on homology. It remains to construct the maps as above.

Consider the end of the resolutions, so we have $f_{-1} : \mathbb{Z} \to \mathbb{Z}$ which is the identity map, and $f_{-2} : 0 \to 0$ is the zero map. Suppose we have already defined f_{n-1}, f_n. We would like to construct f_{n+1}. Thus, $f_n\mathrm{d} : P_{n+1} \to P'_n$, and $\mathrm{d}'(f\mathrm{d}) = f\mathrm{d}\mathrm{d} = 0$. Thus, the map $f\mathrm{d}$ has image contained in $\ker(\mathrm{d}')$. So we can define f_{n+1} as follows:

$$\begin{array}{ccccccc} & & P_{n+1} & \xrightarrow{\mathrm{d}} & P_n & \xrightarrow{\mathrm{d}} & P_{n-1} \\ & {\scriptstyle f_{n+1}}\swarrow & \downarrow{\scriptstyle f_n\mathrm{d}} & & \downarrow{\scriptstyle f_n} & & \downarrow{\scriptstyle f_{n-1}} \\ P'_{n+1} & \twoheadrightarrow & \ker(\mathrm{d}') & \hookrightarrow & P'_n & \xrightarrow[\mathrm{d}']{} & P_{n-1} \end{array}$$

Here, by exactness the map $P'_{n+1} \to \ker(\mathrm{d}')$ is surjective, and so as P'_{n+1} is projective, such a map exists. We can define g_n similarly.

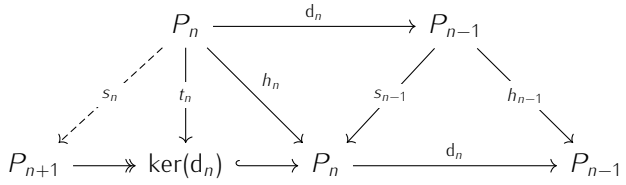
To define s_n, set $h_n = g_n f_n - \mathrm{id}$. This is a map from P_n to itself. In particular, h_n is a chain map, with $h_{-1} = 0$. Set $s_{-1} : \mathbb{Z} \to P_0$ to be the zero map, and note $\mathrm{d}_0 h_0 = h_{-1}\mathrm{d}_0 = 0$, and so $\mathrm{im}(h_0) \subseteq \ker(d_0)$. As before, $\mathrm{d}_1 : P_1 \to \ker(\mathrm{d}_0)$ is surjective, and so we have



Inductively, suppose s_{n-1}, s_{n-2} are already defined, set $t_n = h_n - s_{n-1}\mathrm{d}_n : P_n \to P_n$. We have that

$$\begin{aligned}\mathrm{d}_n t_n &= \mathrm{d}_n h_n - \mathrm{d}_n s_{n-1}\mathrm{d}_n \\ &= h_{n-1}\mathrm{d}_n - (h_{n-1} - s_{n-2}\mathrm{d}_{n-1})\mathrm{d}_n \\ &= s_{n-2}\mathrm{d}_{n-1}\mathrm{d}_n \\ &= 0\end{aligned}$$

So $\mathrm{im}(t_n) \subseteq \ker(\mathrm{d}_n)$. So we have[1]



Define s'_n similarly and we are done. □

Remark 1.35. For any (left) $\mathbb{Z}G$-module N, we can take a projective (resp. free) resolution of N by $\mathbb{Z}G$-modules. Repeating everything we have done, and applying $\mathrm{Hom}_G(\cdot, M)$ gives homology groups, which are called

$$\mathrm{Ext}^n_{\mathbb{Z}G}(N, M)$$

and so

$$\mathrm{H}^n(G, M) = \mathrm{Ext}^n_{\mathbb{Z}}\, G(\mathbb{Z}, M)$$

As above, the Ext-groups are independent of resolution.

2 Low degree cohomology, group extensions

Recall that $\mathrm{H}^0(G, M) = M^G$ is the group of invariants under G, and that a derivation (or a 1-cocycle) is a map $\phi : G \to M$, such that

$$\phi(g_1 g_2) = g_1\phi(g_2) + \phi(g_1)$$

We will see two[2] interpretations of (inner) derivations. Recall that an inner derivation is one of the form $\phi(g) = gm - m$, for some fixed $m \in M$.

Let M be a $\mathbb{Z}G$-module, and consider possible $\mathbb{Z}G$-actions on the abelian group $M \oplus \mathbb{Z}$ of the form

$$g(m, n) = (gm + n\phi(g), n)$$

In this case,

$$g_1 g_2(m, n) = g_1(g_2 m + n\phi(g_2), n) = (g_1 g_2 m + n g_1\phi(g_2) + n\phi(g_1), n)$$

On the other hand,

$$(g_1 g_2)(m, n) = (g_1 g_2 m + n\phi(g_1 g_2), n)$$

[1] THIS DIAGRAM DOES NOT COMMUTE!

[2] The second is on the examples sheet.

These are the same exactly when ϕ is a derivation. In particular, if M is a free $\mathbb{Z}$-module of finite rank, then we get a map $G \to \mathrm{GL}(M)$, by can write this as

$$g \mapsto \begin{pmatrix} \theta_1(g) & \mathrm{d}g \\ 0 & 1 \end{pmatrix}$$

where $\theta_1(g)$ is the action of g on M. In particular, this is a homomorphism of groups if and only if ϕ is a derivation. Moreover, ϕ is an inner derivation if and only if $(-m, 1)$ generates a $\mathbb{Z}G$-submodule which is a trivial module inside $M \oplus \mathbb{Z}$.

Index

$\mathbb{Z}G$-map, 2

augmentation map, 2

chain complex, 3

exact, 3

free $\mathbb{Z}G$ module, 2

$\mathrm{Hom}_G(M, N)$, 2
homology, 3

integral group ring, 1

projective, 4

short exact sequence, 3

trivial $\mathbb{Z}G$ module, 2

$\mathbb{Z}G$ module, 1